VANISHING OF THE POSITIVE COHOMOLOGY OF THE REDUCTION MODULO AN IDEAL SPREADS TO ALL POWER-TORSION AND BOUNDED COMPLEXES WITH POWER-TORSION COHOMOLOGY

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes '(details omitted; hint: represent K by a K-flat complex and compute directly)' and, at the end, 'Details omitted.'; both are kept verbatim. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Derived tensor product as a homotopy colimit over the union of the submodules N[I^n]. used at proof lines 24125-24128. Reduces the arbitrary power-torsion module to one annihilated by a fixed power of I.
- Tor amplitude of a complex and its behaviour modulo a nilpotent ideal. invoked at proof lines 24142-24144. Is the tool that transfers the vanishing from R/I to the square-zero extension.
- Truncation distinguished triangles and induction on the number of nonzero cohomology modules. used at proof lines 24148-24151. Carries the result from modules to bounded complexes.

Source reference. Original lemma 08J5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10838–10838. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 17307: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17307–17307. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let $I \subset R$ be an ideal. Let K be an object of $D(R)$ such that $H^i(K \otimes^{\mathbf{L}}_R R/I) = 0$ for $i > 0$. Then*

(1) *$H^i(K \otimes^{\mathbf{L}}_R R/I^n) = 0$ for all $n \geq 1$ and $i > 0$,*

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(2) $H^i(K \otimes^{\mathbf{L}}_R N) = 0$ *for any* I*-power torsion* R*-module* N *and* $i > 0$*, and*
(3) *for any* $M \in D^b(R)$ *whose cohomology modules* $H^i(M)$ *are* I*-power torsion and* 0 *for* $i > 0$ *we have* $H^i(K \otimes^{\mathbf{L}}_R M) = 0$ *for* $i > 0$*.*

Proof. Proof of (2). We can write $N = \bigcup N[I^n]$. We have $K \otimes^{\mathbf{L}}_R N = \text{hocolim}_n K \otimes^{\mathbf{L}}_R N[I^n]$ as tensor products commute with colimits (details omitted; hint: represent K by a K-flat complex and compute directly). Hence we may assume N is annihilated by I^n. Consider the R-algebra $R' = R/I^n \oplus N$ where N is an ideal of square zero. It suffices to show that the object $K' = K \otimes^{\mathbf{L}}_R R'$ of $D(R')$ has vanishing cohomology in positive degrees. We have a surjection $R' \to R/I$ of R-algebras whose kernel J is nilpotent (any product of n elements in the kernel is zero). We have

$$K \otimes^{\mathbf{L}}_R R/I = (K \otimes^{\mathbf{L}}_R R') \otimes^{\mathbf{L}}_{R'} R/I = K' \otimes^{\mathbf{L}}_{R'} R/I$$

By assumption the complex $K \otimes^{\mathbf{L}}_R R/I$ has tor-amplitude in $[-\infty, 0]$. Thus the conclusion by Lemma 17307.

Part (1) follows trivially from part (2). Part (3) follows from part (2), induction on the number of nonzero cohomology modules of M, and the distinguished triangles of truncation from Derived Categories, Remark 08J5. Details omitted. □